

Tanvi Hernandez

Embedded Systems & IoT Firmware Engineer • 9 Years Experience • MERIQ Index: 94% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, I2C/SPI/UART, Python, FreeRTOS, Embedded Linux, ARM Cortex, PCB Debugging

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 32%.
- Led a squad of 8 engineers delivering scalable microservices processing 17M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 52%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2017 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using Python saving \$39k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [C/C++, I2C/SPI/UART, Python, FreeRTOS]
Engineered end-to-end production solution leveraging C/C++ and I2C/SPI/UART with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, FreeRTOS, Embedded Linux, ARM Cortex]
Architected high-resilience system integrating Python with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Robotics and Autonomous Systems

IIT Delhi — 2013 - 2017 (Score: 3.69 / 4.0)